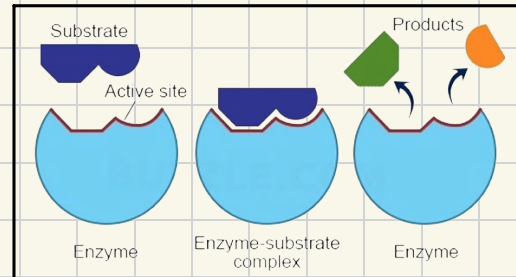


enzymes

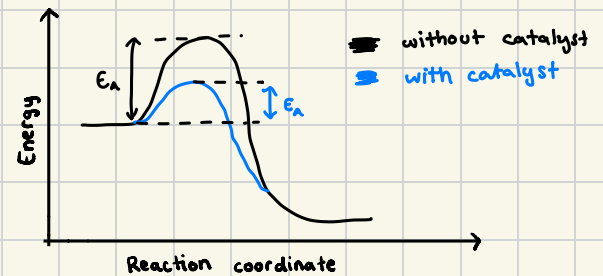
Enzymes:

- biological catalysts made of globular proteins
- catalyse intracellular & extracellular reactions
e.g. catalase inside liver cells or trypsin in the small intestine
- the active site is specific & unique in shape due to the specific folding & bonding in the tertiary structure of the protein
- due to this, the enzymes can only attach to substrates that are complementary in shape



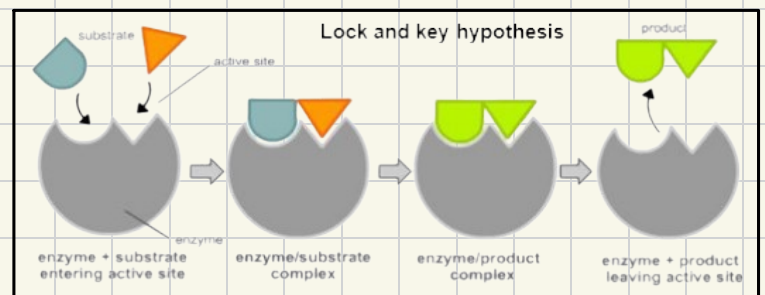
Activation energy:

- all reactions require a certain amount of energy before they occur, known as activation energy (E_a)
- when enzymes attach to the substrate, they can lower the E_a needed for the reaction to occur & thus speed up the reaction.



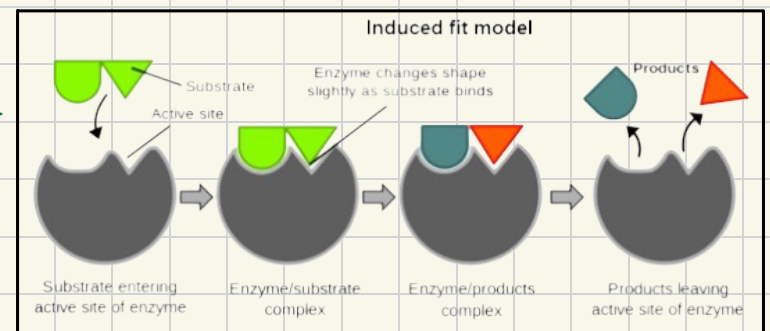
Lock-and-key hypothesis

- this model suggests that the enzyme is like a lock & the substrate is like a key
- it's thought that the active site is a fixed shape and that due to random collision, an enzyme-substrate complex is formed
- the charged groups within the active site are thought to distort the substrate thus lowering E_a



Induced fit hypothesis

- this model suggests that the enzyme is like a glove & the substrate is like your hand
- it's thought that the active site is induced, or slightly changes shape, to mould around the substrate
- when the enzyme-substrate complex occurs, it puts strain on the bonds & thus lowers E_a



Investigating enzyme-catalysed reactions

• catalase reaction monitoring

- ↳ catalase is an enzyme that catalyses the breakdown of hydrogen peroxide (substrate) into water & oxygen
- ↳ to investigate the progress of this reaction, we can measure the rate of formation of products, namely oxygen, over time
- ↳ this can be done by collecting the oxygen gas evolved during the reaction in a calibrated measuring cylinder & recording the volume at regular time intervals
- ↳ alternatively, the rate of oxygen production can be indirectly measured by using a gas pressure sensor connected to a data logger, which records changes in pressure
- ↳ by plotting the volume or pressure of oxygen produced against time on a graph, we can visualise the rate of the catalase-catalysed reaction

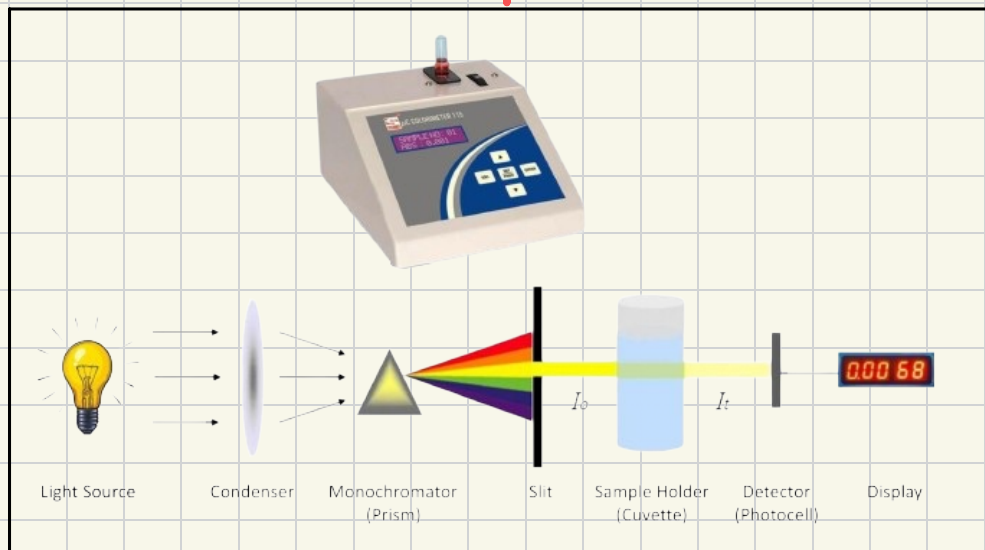
• amylase reaction monitoring

- ↳ amylase is an enzyme that catalyses the hydrolysis of starch into simpler sugars
- ↳ to investigate the progress of this reaction, we can measure the rate of disappearance of the starch substrate over time
- ↳ this can be done by taking samples of the reaction mixture at regular intervals & test for the presence of starch using an iodine solution
- ↳ as the starch is broken down into simpler sugars, the iodine solution will change from blue-black to orange/brown, indicating the depletion of starch
- ↳ by quantifying the intensity of the remaining color or lack, using a **colorimeter**, the rate of starch disappearance can be determined

→ an instrument used to measure the absorbance/transmittance of light by a solution

→ it works on the principle that certain substances absorb light at specific wavelengths, resulting in a change of intensity of the transmitted light

→ enzyme-catalysed reactions often involve color changes due to the formation of products or loss of substrates



Factors affecting enzyme-catalysed reactions

① temperature

- if temp. is too low, there is insufficient kinetic energy for successful collisions
- if temp. is too high, the enzymes denature; the tertiary structure alters due to bonds breaking and active site changes shape thus enzyme-substrate complexes cannot be formed

② pH

- too high or low pH interferes with the charges in the amino acids in the active site which causes bonds to break & tertiary structure alters thus enzyme denatures

* all enzymes have different optimal temp. & pH that they work at

③ enzyme concentration

- low enzyme conc. will result in a lower rate of reaction
- increasing enzyme conc. will increase the rate of reaction as enzyme-substrate complexes are more likely to form. This would happen until the rate of reaction plateaus due to insufficient substrate present to bind with the large number of enzymes

④ substrate concentration

- low substrate conc. will result in a lower rate of reaction as there will be fewer collisions between the enzyme & substrate
- increasing substrate conc. will increase the rate of reaction as enzyme-substrate complexes are more likely to form. This would happen until the rate of reaction plateaus due to the fact that all the enzyme active sites would be in use

⑤ inhibitor concentration

competitive inhibitors

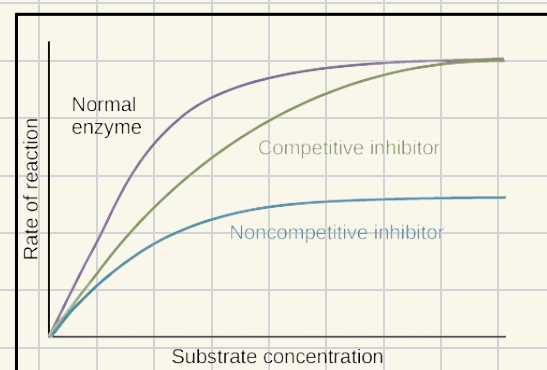
- complementary in shape to active site thus have the ability to bind to it
- this prevents the substrate from binding & enzyme-inhibitor complexes form instead of enzyme-substrate complexes.
- most competitive inhibitors are reversible so if a high enough conc. of substrate is added, the substrate can knock out the inhibitor & increase the rate of reaction

non-competitive inhibitors

- bind to the enzymes allosteric site, away from the active site
- this causes the active site to change shape and thus the substrate can no longer bind, regardless of how much is added
- as a result, fewer enzyme-substrate complexes form & the rate of reaction is much lower

* the products of some reactions are reversible inhibitors for the enzymes

* end-product inhibition enables the reactions to be controlled, to prevent overheating of cell or wastage of resources



Max. rate of reaction :

- V_{max} refers to the max. rate of reaction at which an enzyme catalyses a reaction
- it's reached when the active sites of all enzymes are fully saturated & all enzymes are working at max. capacity, so even if substrate conc. is increased, the rate of reaction wouldn't increase
- V_{max} is a key measure to characterise enzyme kinetics & is determined experimentally by plotting reaction rate against substrate concentration until plateau is reached

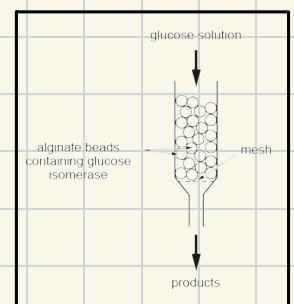
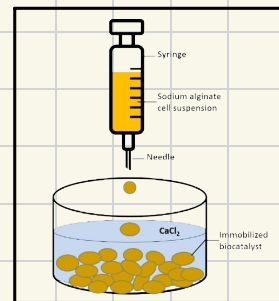
Michaelis - Menten constant (K_m) :

- K_m is a measure of the affinity of an enzyme for its substrate. It represents the substrate concentration at which the reaction rate is equal to half of V_{max}
- it can be thought of as the substrate conc. required to achieve half-maximal velocity
- enzymes with lower K_m values have a higher affinity for their substrates, as they reach half-maximal velocity at lower substrate conc.
- the Michaelis-Menten equation, which describes the relationship between reaction rate, substrate conc., V_{max} & V_{min}
- K_m values can be used to compare the affinity of different enzymes for their substrates
- enzymes with lower K_m values exhibit greater substrate affinity & are more efficient at converting substrate into product, even at low substrate conc., while enzymes with higher K_m values have lower substrate affinity & require more substrate conc. for maximal velocity.

Enzymes immobilised in alginate vs. in free solution

How to encapsulate enzymes in alginate beads?

- 1) mix enzymes with sodium alginate
- 2) add drop by drop into calcium chloride
- 3) pack them into a column & run it through
- 4) collect products from bottom



advantages of using immobilised enzymes

- + can easily reuse & recover the enzymes
- + longer shelf-life of the enzymes
- + less purification / downstream processing needed because product is enzyme-free
- + allow continuous production & can obtain more product per unit time
- + reduces end product inhibition
- + enzymes more tolerant of pH changes & are thermostable